



Systematic Evaluation of Sigmoid, ReLU,
and Tanh

Neural Network Activation Functions Comparison MNIST Digit Classification Analysis

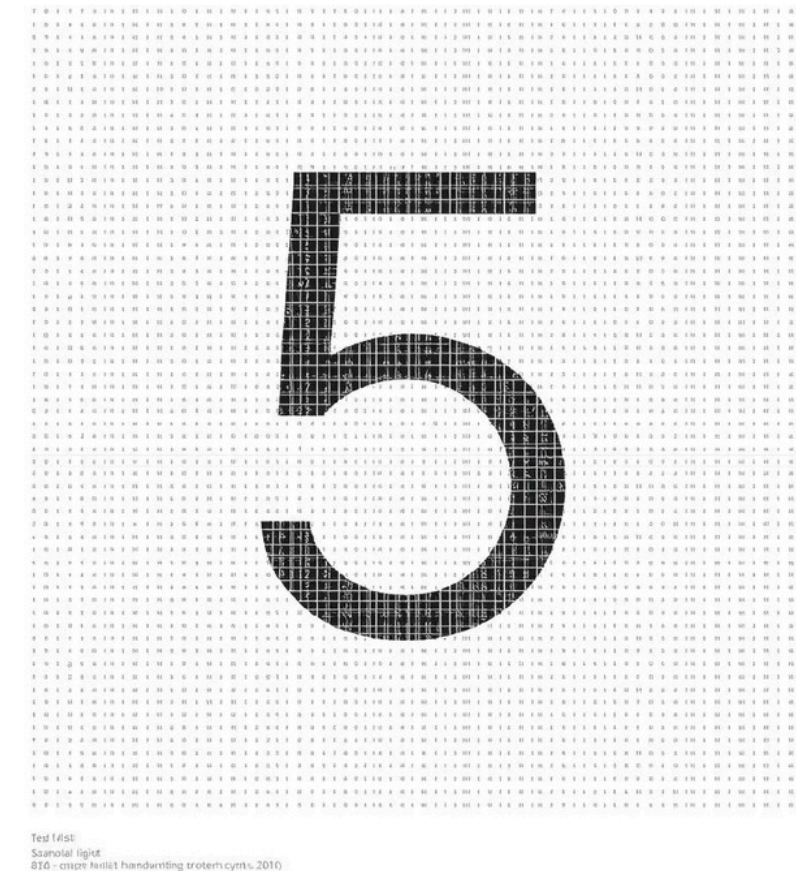


Introduction & Study Context

Systematic comparison of activation functions on MNIST digit classification. Testing Sigmoid, ReLU, and Tanh activations across small vs large network architectures to identify optimal configurations for digit recognition.

Data Preprocessing

MNIST dataset contains 60,000 training and 10,000 test images. Each 28×28 pixel image is flattened to a 784-dimensional vector. Pixel values are normalized from 0–255 to 0–1 range. Labels are converted to one-hot encoding across 10 digit classes.



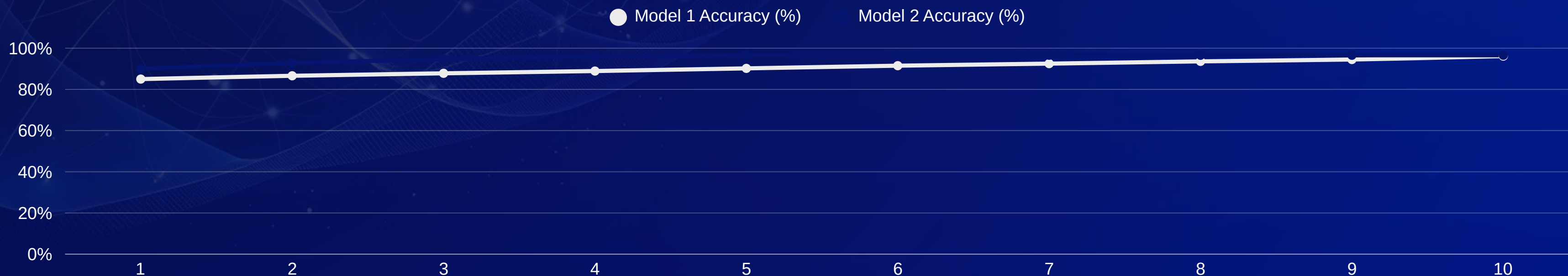
Model Architectures Tested

Sigmoid: $32 \rightarrow 16 \rightarrow 10$ (10 epochs) & $128 \rightarrow 64 \rightarrow 10$ (5 epochs, $lr=0.01$). ReLU: $64 \rightarrow 32 \rightarrow 10$ (10 epochs) & $128 \rightarrow 64 \rightarrow 10$ (6 epochs, $lr=0.02$). Tanh: $64 \rightarrow 32 \rightarrow 10$ (10 epochs) & $128 \rightarrow 64 \rightarrow 10$ (4 epochs, $lr=0.04$). All use Adam optimizer with categorical crossentropy loss.



Sigmoid Models Performance

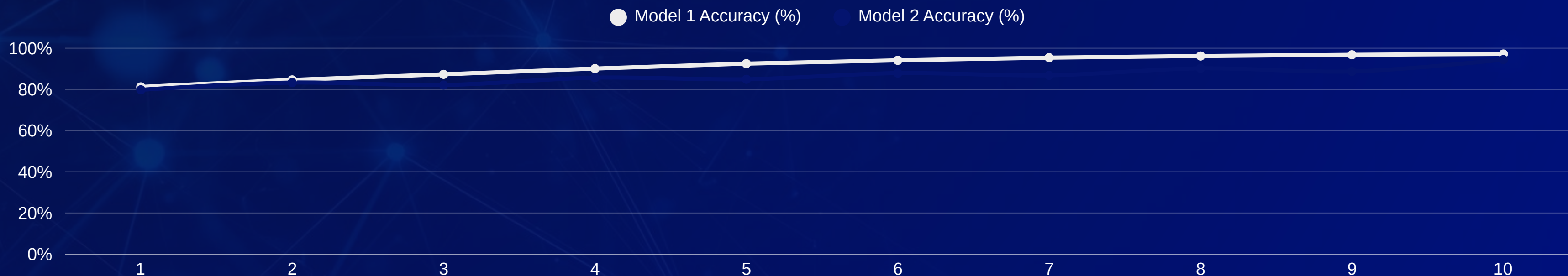
Model 1 Accuracy: 96.15%, loss: 0.1304
Model 2 Accuracy: 96.56%, loss: 0.1299
Model 2 converges faster due to larger network capacity (128→64→10 vs 32→16→10 neurons).



Source: AI-generated data. Replace or verify before use.

ReLU Models Performance

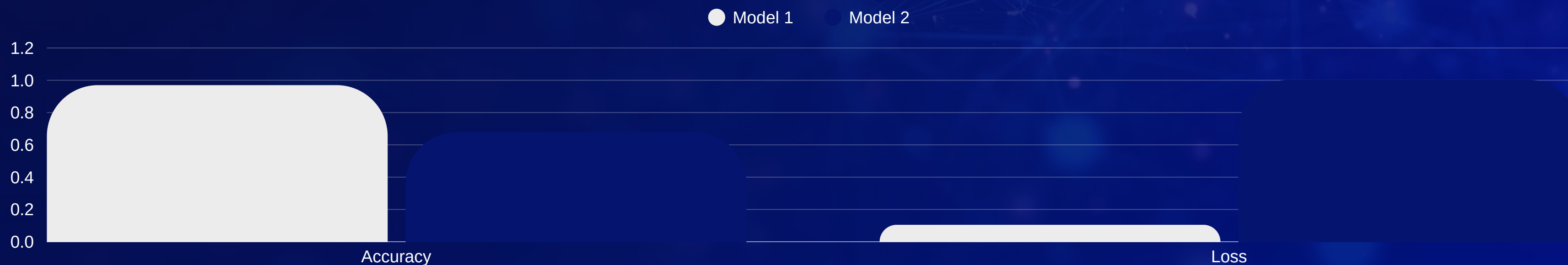
Model 1 achieves 97.17% accuracy with loss of 0.1095, representing the best overall performance across all activation functions tested. Model 2 reaches 94.44% accuracy but exhibits unstable training behavior with higher loss of 0.2449. The smaller architecture demonstrates superior stability and convergence.



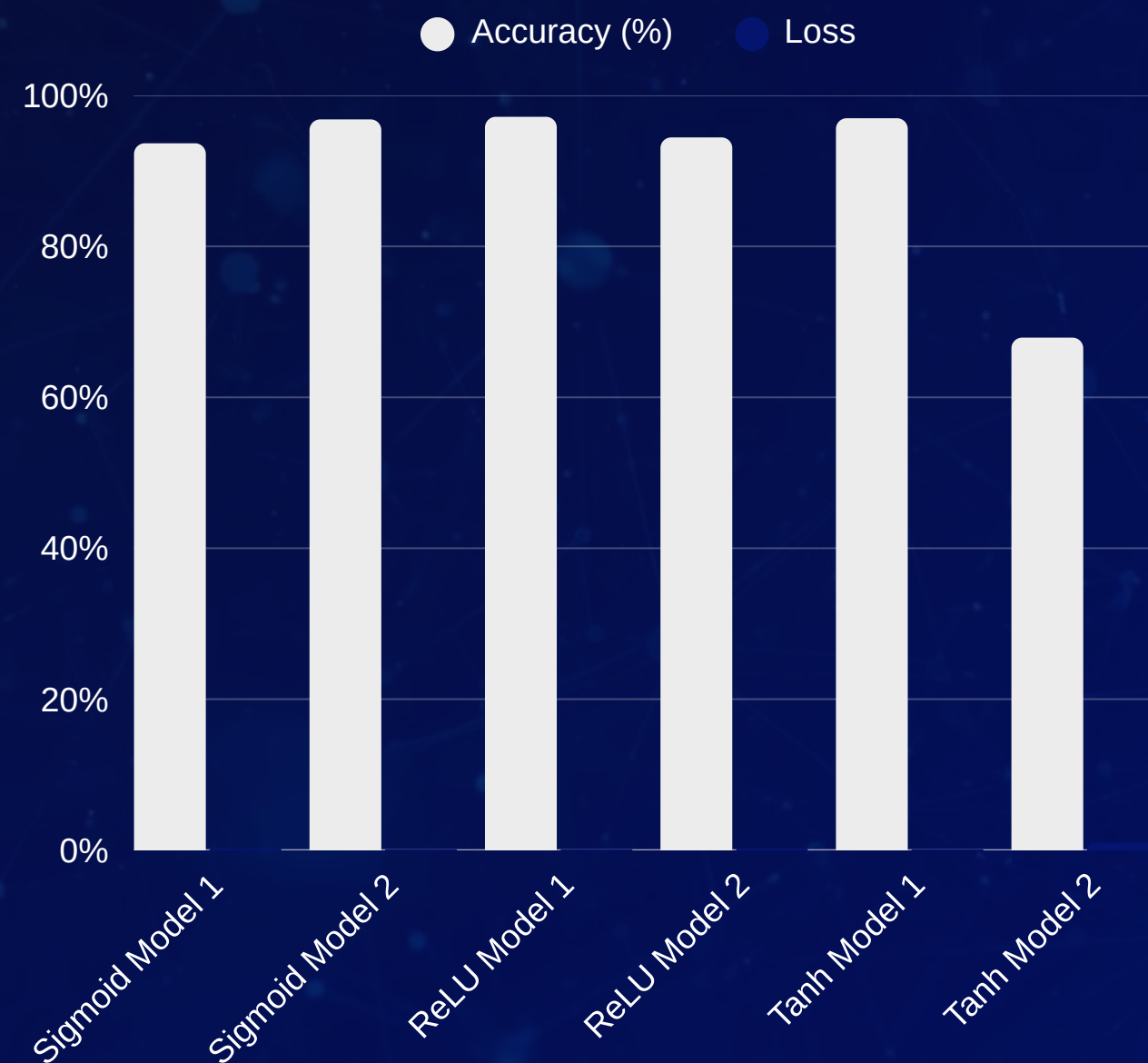
Source: AI-generated data. Replace or verify before use.

Tanh Models Performance

Model 1 achieves 96.98% accuracy with stable training and low loss of 0.1050. Model 2 suffers catastrophic failure at just 67.91% accuracy with severe instability and loss of 1.0030, demonstrating the risks of oversized networks without proper regularization.



Source: AI-generated data. Replace or verify before use.



Source: AI-generated data. Replace or verify before use.

Final Comparison Summary

ReLU Model 1 achieves the highest accuracy at 97.17%, while Tanh Model 2 catastrophically fails at 67.91%. Moderate network architectures consistently outperform larger networks across all activation functions.

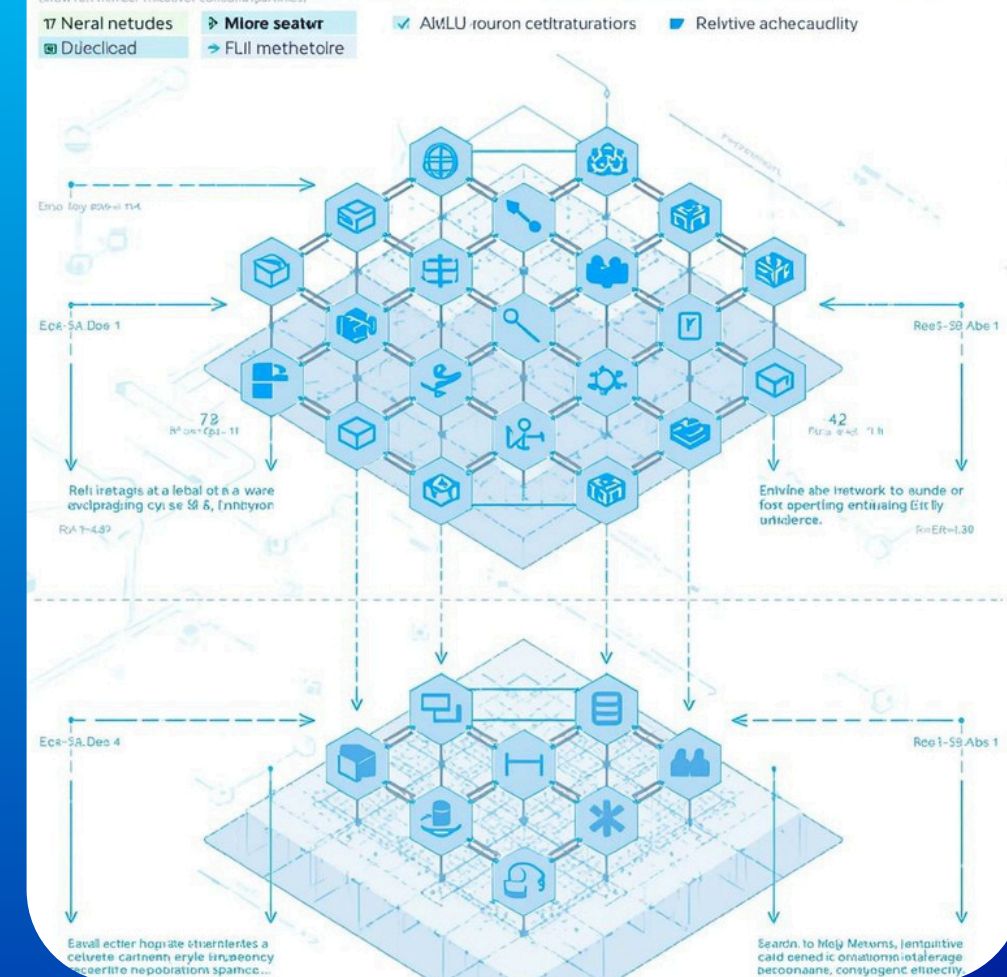
Practical Recommendations

Use ReLU activation for hidden layers in image classification tasks. Prefer moderate network sizes (e.g., 64→32→10 neurons) over larger architectures. Normalize input data to 0-1 range. Monitor validation accuracy to detect overfitting early. Employ early stopping to prevent overtraining and balance learning rate with network capacity.

Neural Network architecture

inserirur vidufons infrivatiois

A Indral sode for-neural network, Drē-Rel-retivared condations: lajines and-emvbal activevture, isuinesss perductzied, totes with lis d ilul-desamaris marnally, ended network. refirms fteet lady. J alll tipped networs, layer dsignam oeroereany ave, dettural cerifications att-now dnerore and pad. leam vnyorim theacroem. lentilit, mal amrsil:ciat asemearbe ared octutioesvd-in loselty.



Conclusion & Key Takeaway

"In MNIST digit classification, smarter architecture beats sheer size every time." Moderate-sized networks with ReLU activation deliver the best accuracy at 97.17%, proving that thoughtful design outperforms brute-force scaling.

Thank you for your attention

•

Questions & Discussion